Line types, dimensioning & scale

GRAPHICS WITH MR KELLY

Higher · Drawing & spatial awareness · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name and draw outline, projection, dimension, hidden detail, centre, cutting plane and fold lines
- ☐ apply chain, parallel, radial, diameter, angular, square, A/F and A/C dimensioning
- ☐ spot British Standards errors on an orthographic drawing
- ☐ use and calculate with scales
- ☐ explain and apply tolerances and limits

I DO — Worked example

Read how a full-mark answer is built

A production drawing of a coat hook contains several British Standards errors.

1 Describe four errors that commonly appear on orthographic drawings and how each should be corrected. (4 marks)

Full-mark answer each ✓ is one mark

The Ø20 on the fillet should be a radius, R10 ✓. The arrowhead on the Ø20 doesn't touch the arc and must touch it ✓. Centre lines are missing from the hole in the plan and should be added ✓. The plan is drawn below the elevation; in third angle it must be above ✓.

Why answers lose marks

"Wrong dimension, missing lines, wrong view, bad arrows."

0–1 marks: errors are listed without saying exactly what is wrong or how to correct it.

WE DO — Guided practice

Use the starters and word bank to help you

A linear dimension on a drawing reads □25.

2 Explain the purpose of the symbol. (1 mark)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

feature · cross-section · square

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Dimensions placed end to end are called...
☐ Parallel ☐ Chain ☐ Datum ☐ Angular

b) Dimension figures should be readable from...
☐ Any direction ☐ The bottom or right of the sheet ☐ The top ☐ The left only

c) Which avoids the build-up of errors?
☐ Chain dimensioning ☐ Parallel (datum) dimensioning ☐ Leader lines ☐ Hidden detail

d) The symbol □ before a dimension means...
☐ Cubed ☐ Square section ☐ Area ☐ Approximate

4 Key terms: write the letter of the matching description next to each term.

1. **Centre line**	____	**A** All sizes from one datum; avoids accumulated error.
2. **Cutting plane line**	____	**B** Dimensions end to end; shows feature-to-feature sizes but errors accumulate.
3. **Fold line**	____	**C** The feature is square in section.
4. **Chain dimensioning**	____	**D** Thin chain line: long dash, short dash.
5. **Parallel (datum) dimensioning**	____	**E** Thin chain, thick at the ends, with arrows and letters.
6. **□20 on a dimension**	____	**F** Thin chain line with two short dashes, used on surface developments.

In a detail view of a small clip, a feature that is really 12 mm long measures 60 mm on the drawing.

5 State the scale of the detail view. (1 mark)

6 State two methods of showing a tolerance on a production drawing. (2 marks)

An enlarged detail of a surface development for a carton shows three different line types, labelled X, Y and Z.

7 State the name of each line type: X runs along a crease, Y is thick and continuous around the edge, Z runs through the centre of a circular window. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?